

EMPLOYEE ATTRITION ANALYSIS PROJECT

Siddharth Nair

OBJECTIVE

Analyse employee attrition data and identify factors affecting employee turnover. Present findings through visualizations, insights, and recommendations.

PROCEDURE

I. Create data dictionary

COLUMN NAME	DATA TYPE	RANGE
Age	int64	18 – 60
Attrition	object	Yes / No
BusinessTravel	object	'Travel_Rarely', 'Travel_Frequently', 'Non-Travel'
DailyRate	int64	102 – 1499
Department	object	'Sales', 'Research & Development', 'Human Resources'
DistanceFromHome	int64	1 – 29
Education	int64	1 – 5
EducationField	object	'Life Sciences', 'Other', 'Medical', 'Marketing', 'Technical Degree', 'Human Resources'
EmployeeCount	int64	1
EmployeeNumber	int64	1 – 2068
EnvironmentSatisfaction	int64	1 – 4
Gender	object	Male / Female
HourlyRate	int64	30 – 100
JobInvolvement	int64	1 – 4
JobLevel	int64	1 – 5
JobRole	object	'Sales Executive', 'Research Scientist', 'Laboratory Technician', 'Manufacturing Director', 'Healthcare Representative', 'Manager', 'Sales Representative', 'Research Director', 'Human Resources'
JobSatisfaction	int64	1 – 4
MaritalStatus	object	'Single', 'Married', 'Divorced'
MonthlyIncome	int64	1009 – 19999
MonthlyRate	int64	2094 – 26999

NumCompaniesWorked	int64	0 – 9
Over18	object	Yes
OverTime	object	Yes / No
PercentSalaryHike	int64	11 – 25
PerformanceRating	int64	3 – 4
RelationshipSatisfaction	int64	1 – 4
StandardHours	int64	80
StockOptionLevel	int64	0 – 3
TotalWorkingYears	int64	0 – 40
TrainingTimesLastYear	int64	0 – 6
WorkLifeBalance	int64	1 – 4
YearsAtCompany	int64	0 – 40
YearsInCurrentRole	int64	0 – 18
YearsSinceLastPromotion	int64	0 – 15
YearsWithCurrManager	int64	0 – 17

2. Clean the data

The employee attrition dataset was loaded and its structure, size, data types, missing values, and duplicate records were inspected. I checked important categorical and numerical values for consistency and validity. I removed unnecessary columns such as *EmployeeCount*, *Over18*, *StandardHours*, and *EmployeeNumber*. Finally, I verified the cleaned dataset and saved it as a new CSV file for further EDA and analysis.

3. Identify numerical and categorical columns.

The Numeric Columns are:

Age, DailyRate, DistanceFromHome, Education, EnvironmentSatisfaction, HourlyRate, JobInvolvement, JobLevel, JobSatisfaction, MonthlyIncome, MonthlyRate, NumCompaniesWorked, PercentSalaryHike, PerformanceRating, RelationshipSatisfaction, StockOptionLevel, TotalWorkingYears, TrainingTimesLastYear, WorkLifeBalance, YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion and YearsWithCurrManager.

The Categorical Columns are:

Attrition, BusinessTravel, Department, EducationField, Gender, JobRole, MaritalStatus and OverTime.

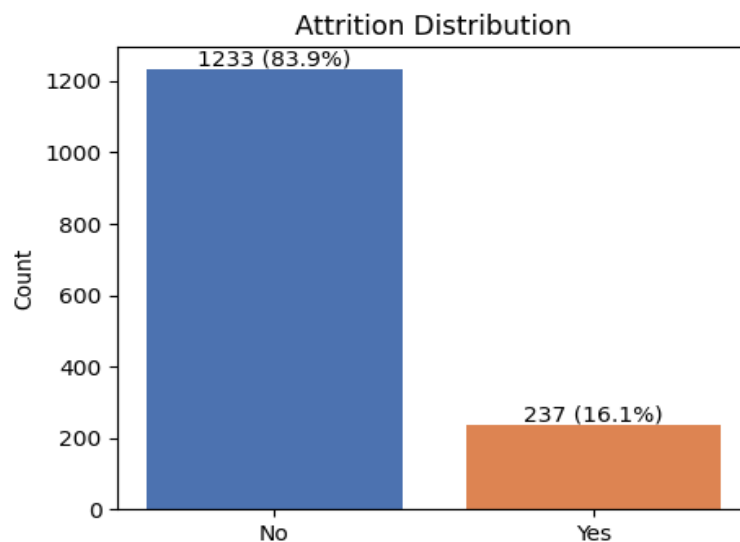
4. Conduct exploratory data analysis (EDA)

Exploratory Data Analysis (EDA) is a critical initial phase in data science and machine learning where analysts investigate datasets to summarize main characteristics, often using visual methods, before building formal models. IT helps us identify patterns, detect anomalies, and uncover relationships between variables without preconceived hypotheses.

Analysing the dataset, we find that there is not singular factor that strongly correlates to the target feature (Attrition). Instead, we observe a cluster of overlapping traits point inverse relationship with target, most notably working overtime, lesser age, marital status as single, earlier in one's career, and in lower-paying or lower-level roles. On the contrary, frequent business travel correlated with higher attrition. When considering the job roles solely, Sales Representatives leave at the highest rate while Research Directors and Managers were the most stable, largely because those senior positions naturally overlap with higher tenure and income.

▼ 1. Attrition Distribution

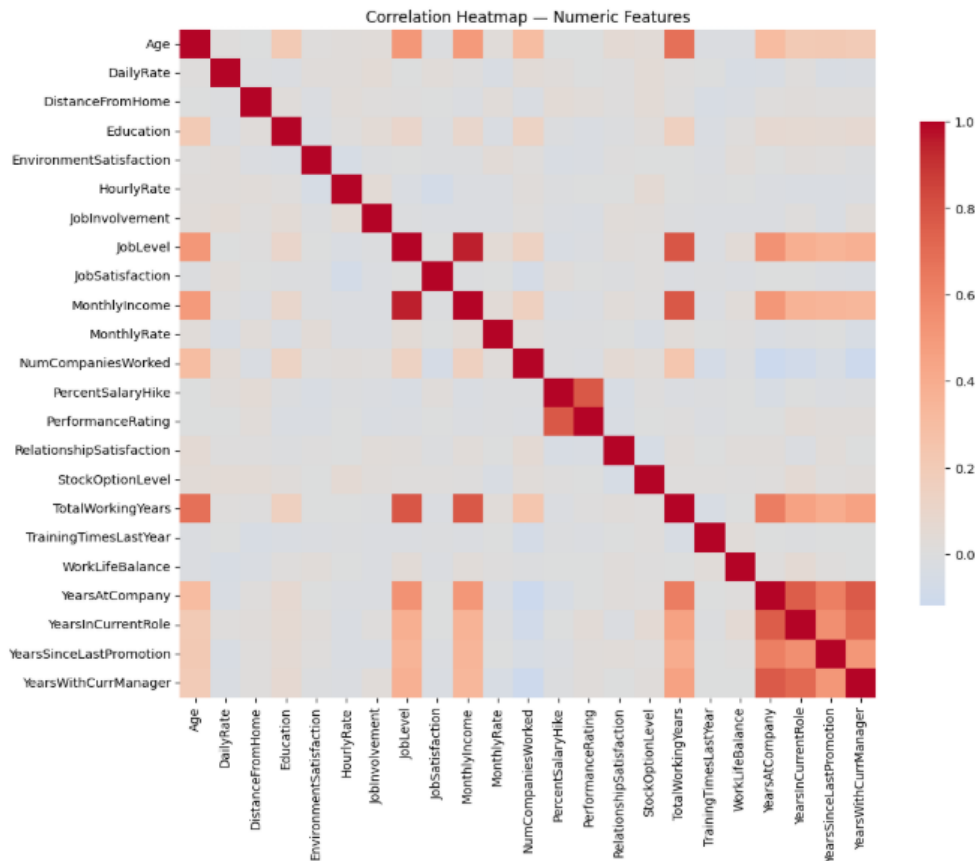
```
i]: counts = df['Attrition'].value_counts()
fig, ax = plt.subplots(figsize=(5,4))
ax.bar(counts.index, counts.values, color=['#4C72B0','#DD8452'])
for i, v in enumerate(counts.values):
    ax.text(i, v + 10, f"{v} ({v/len(df)*100:.1f}%)", ha='center')
ax.set_title('Attrition Distribution')
ax.set_ylabel('Count')
plt.show()
```



2. Correlation Structure

```
[4]: num_df = df.select_dtypes(include=[np.number])
     corr = num_df.corr()

     fig, ax = plt.subplots(figsize=(12,10))
     sns.heatmap(corr, cmap='coolwarm', center=0, ax=ax, square=True, cbar_kws={'shrink':0.7})
     ax.set_title('Correlation Heatmap - Numeric Features')
     plt.show()
```



3. Numeric features vs. Attrition

```
[6]: df['Attrition_bin'] = (df['Attrition'] == 'Yes').astype(int)
     corrs = num_df.assign(Attrition_bin=df['Attrition_bin']).corr()['Attrition_bin'].drop('Attrition_bin').sort_values()
     corrs
```

```
[6]: TotalWorkingYears    -0.171063
     JobLevel            -0.169105
     YearsInCurrentRole  -0.160545
     MonthlyIncome       -0.159840
     Age                 -0.159205
     YearsWithCurrManager -0.156199
     StockOptionLevel    -0.137145
     YearsAtCompany      -0.134392
     JobInvolvement      -0.130016
     JobSatisfaction     -0.103481
     EnvironmentSatisfaction -0.103369
     WorkLifeBalance     -0.063939
     TrainingTimesLastYear -0.059478
     DailyRate           -0.056652
     RelationshipSatisfaction -0.045872
     YearsSinceLastPromotion -0.033019
     Education           -0.031373
     PercentSalaryHike   -0.013478
     HourlyRate          -0.006846
     PerformanceRating    0.002889
     MonthlyRate         0.015170
     NumCompaniesWorked  0.043494
     DistanceFromHome    0.077924
     Name: Attrition_bin, dtype: float64
```

Note: lower TotalWorkingYears, lower JobLevel, less time InCurrentRole, lower MonthlyIncome, and younger Age are all associated with higher attrition.

```

fig, axes = plt.subplots(2, 2, figsize=(11,9))
sns.boxplot(data=df, x='Attrition', y='Age', ax=axes[0,0], hue='Attrition', palette=['#4C72B8','#D08452'], legend=False)
axes[0,0].set_title('Age by Attrition')

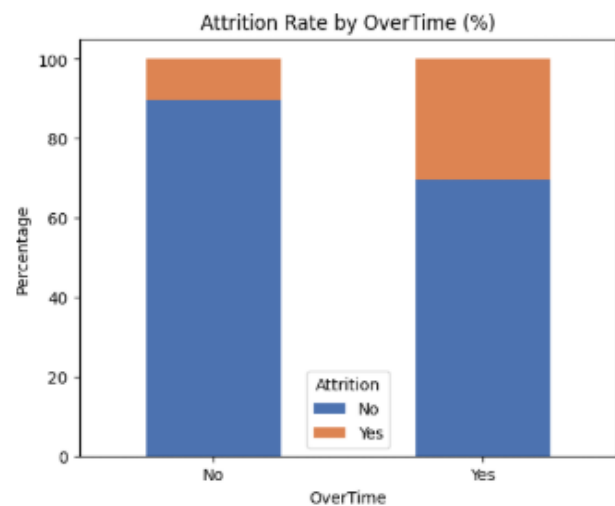
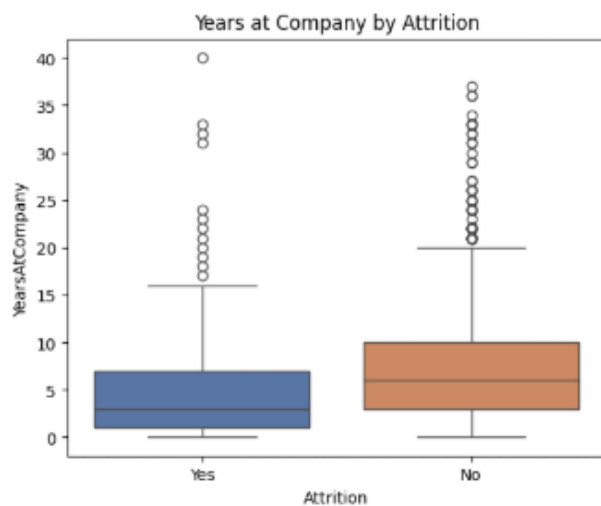
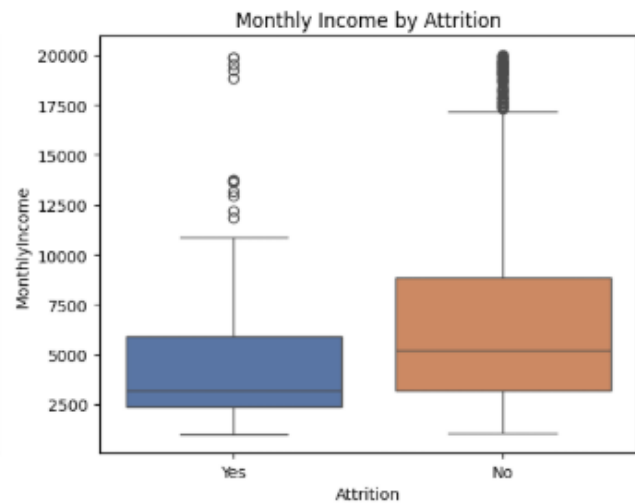
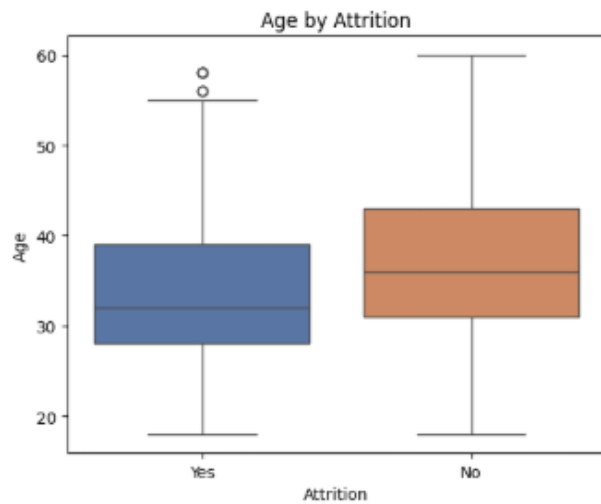
sns.boxplot(data=df, x='Attrition', y='MonthlyIncome', ax=axes[0,1], hue='Attrition', palette=['#4C72B8','#D08452'], legend=False)
axes[0,1].set_title('Monthly Income by Attrition')

sns.boxplot(data=df, x='Attrition', y='YearsAtCompany', ax=axes[1,0], hue='Attrition', palette=['#4C72B8','#D08452'], legend=False)
axes[1,0].set_title('Years at Company by Attrition')

ct = pd.crosstab(df['OverTime'], df['Attrition'], normalize='index') * 100
ct.plot(kind='bar', stacked=True, ax=axes[1,1], color=['#4C72B8','#D08452'])
axes[1,1].set_title('Attrition Rate by OverTime (%)')
axes[1,1].set_ylabel('Percentage')
axes[1,1].tick_params(axis='x', rotation=0)

plt.tight_layout()
plt.show()

```



4. Categorical / behavioral drivers

```
[8]: for col in ['OverTime', 'MaritalStatus', 'BusinessTravel', 'Department']:
      print(f"\nAttrition rate by {col}:")
      print((df.groupby(col)['Attrition_bin'].mean() * 100).round(1).sort_values(ascending=False))
```

```
Attrition rate by OverTime:
OverTime
Yes      30.5
No       10.4
Name: Attrition_bin, dtype: float64
```

```
Attrition rate by MaritalStatus:
MaritalStatus
Single      25.5
Married     12.5
Divorced    10.1
Name: Attrition_bin, dtype: float64
```

```
Attrition rate by BusinessTravel:
BusinessTravel
Travel_Frequently  24.9
Travel_Rarely      15.0
Non-Travel         8.0
Name: Attrition_bin, dtype: float64
```

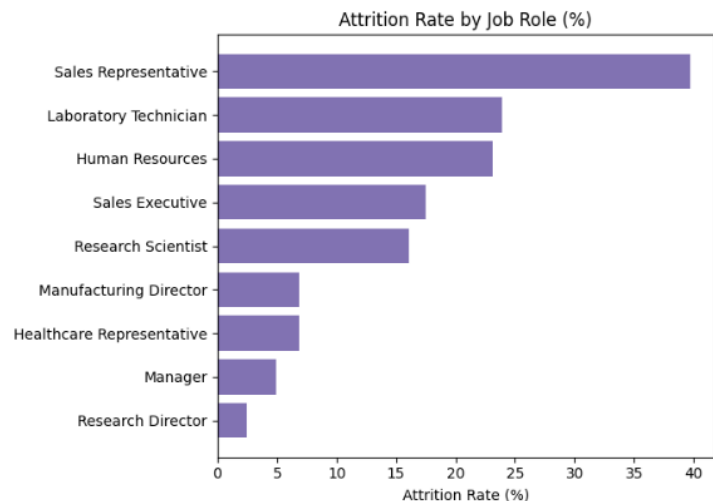
```
Attrition rate by Department:
Department
Sales                20.6
Human Resources      19.0
Research & Development 13.8
Name: Attrition_bin, dtype: float64
```

5. Attrition by Job Role

```
[9]: role_attr = df.groupby('JobRole')['Attrition_bin'].mean().mul(100).sort_values()

fig, ax = plt.subplots(figsize=(7,5))
ax.barh(role_attr.index, role_attr.values, color='#8172B2')
ax.set_title('Attrition Rate by Job Role (%)')
ax.set_xlabel('Attrition Rate (%)')
plt.tight_layout()
plt.show()

role_attr.sort_values(ascending=False)
```



```
[9]: JobRole
Sales Representative      39.759036
Laboratory Technician    23.938224
Human Resources           23.076923
Sales Executive           17.484663
Research Scientist        16.095890
Manufacturing Director    6.896552
Healthcare Representative  6.870229
Manager                   4.901961
Research Director         2.500000
Name: Attrition_bin, dtype: float64
```

5. Calculate key HR metrics (KPIs)

HR KPIs (Human Resources Key Performance Indicators) are strategic, quantifiable metrics used to measure the effectiveness of HR initiatives and their direct contribution to an organization's broader business goals. Unlike general HR metrics that track activity volume (such as total training hours or headcount), KPIs are focused on outcomes that drive performance, such as employee retention rates, quality of hire, and employee productivity.

$$\begin{aligned} \text{i. Turnover Rate} &= \frac{\text{Number of Employees who left}}{\text{Total Number of Employees}} \times 100\% \\ &= \frac{237}{1470} \times 100\% \approx 16.1\% \end{aligned}$$

$$\text{ii. Average Tenure} = \frac{\text{Total Time in Role}}{\text{Total Employees in Role}} \times 100\%$$

JOB ROLE	AVERAGE TIME (IN YEARS)
Healthcare Representative	14.068702
Human Resources	8.173077
Laboratory Technician	7.656371
Manager	24.549020
Manufacturing Director	12.786207
Research Director	21.400000
Research Scientist	7.715753
Sales Executive	11.101227
Sales Representative	4.674699

iii. Average Monthly Income per Department

DEPARTMENT	AVERAGE SALARY
Human Resources	6654.507937
Research & Development	6281.252862
Sales	6959.172646

$$\text{iv. Overtime Rate} = \frac{\text{Number of Employees Working Overtime}}{\text{Total Number of Employees}} \times 100\%$$

$$= \frac{416}{1470} \times 100\% \approx 28.3\%$$

$$\text{v. Average Performance Rate} = \frac{\text{Sum of Performance Ratings}}{\text{Total Number of Employees}}$$

$$= \frac{4636}{1470} \approx 3.15$$

6. Analyse attrition across different employee segments

Attrition % split by Department

Attrition	No	Yes	Count
Department			
Sales	79.4	20.6	446
Human Resources	81.0	19.0	63
Research & Development	86.2	13.8	961

Attrition % split by JobRole

Attrition	No	Yes	Count
JobRole			
Sales Representative	60.2	39.8	83
Laboratory Technician	76.1	23.9	259
Human Resources	76.9	23.1	52
Sales Executive	82.5	17.5	326
Research Scientist	83.9	16.1	292
Healthcare Representative	93.1	6.9	131
Manufacturing Director	93.1	6.9	145
Manager	95.1	4.9	102
Research Director	97.5	2.5	80

Attrition % split by Gender

Attrition	No	Yes	Count
Gender			
Male	83.0	17.0	882
Female	85.2	14.8	588

Attrition % split by MaritalStatus

Attrition	No	Yes	Count
MaritalStatus			
Single	74.5	25.5	470
Married	87.5	12.5	673
Divorced	89.9	10.1	327

Attrition % split by OverTime

Attrition	No	Yes	Count
OverTime			
Yes	69.5	30.5	416
No	89.6	10.4	1054

Attrition % split by BusinessTravel

Attrition	No	Yes	Count
BusinessTravel			
Travel_Frequently	75.1	24.9	277
Travel_Rarely	85.0	15.0	1043
Non-Travel	92.0	8.0	150

Attrition % split by EducationField

Attrition	No	Yes	Count
EducationField			
Human Resources	74.1	25.9	27
Technical Degree	75.8	24.2	132
Marketing	78.0	22.0	159
Life Sciences	85.3	14.7	606
Medical	86.4	13.6	464
Other	86.6	13.4	82

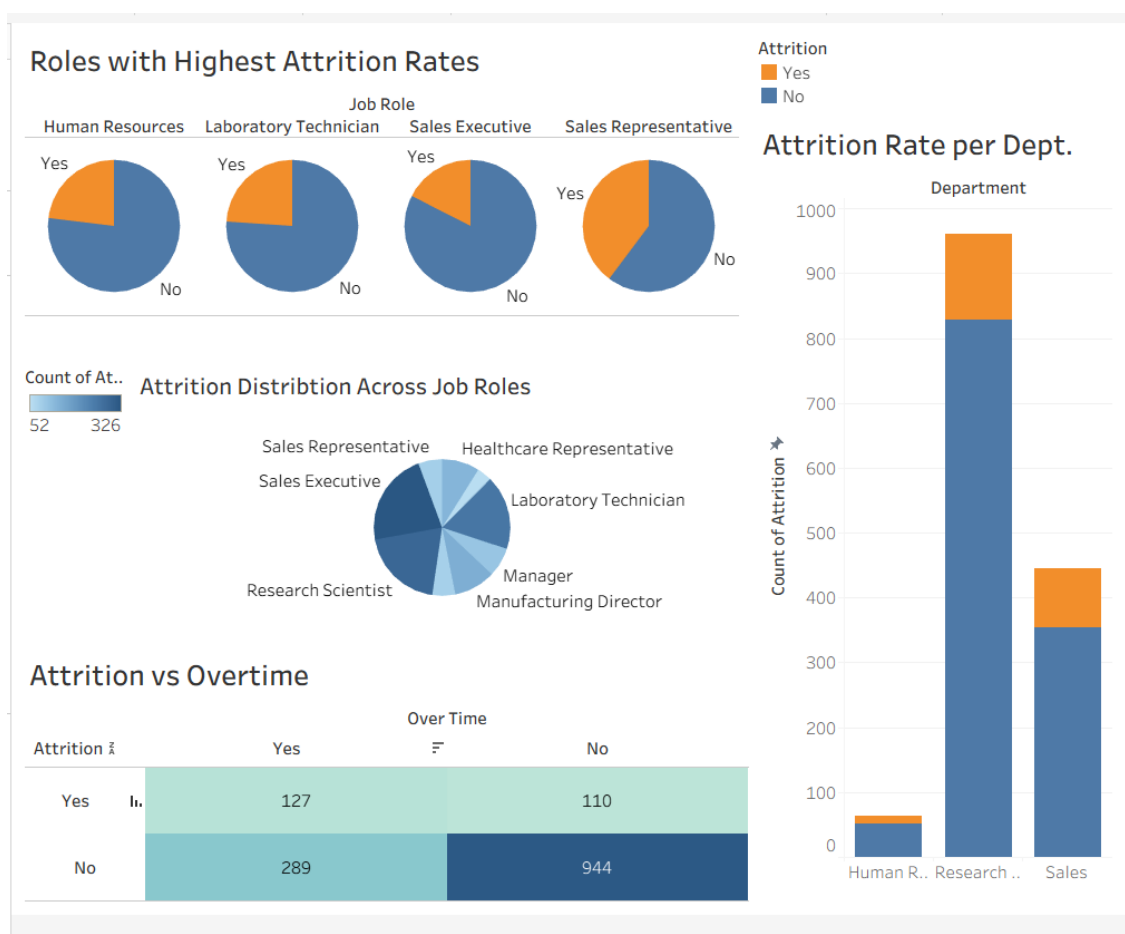
Attrition % split by JobLevel

Attrition	No	Yes	Count
JobLevel			
1	73.7	26.3	543
3	85.3	14.7	218
2	90.3	9.7	534
5	92.8	7.2	69
4	95.3	4.7	106

7. Identify patterns related to salary, overtime, job satisfaction, and work-life balance.

Analysing salary, overtime, job satisfaction, and work-life balance show us that attrition is not driven by any single factor but by how these factors compound. Low earners face the highest risk, with 29.3% attrition in the bottom income quartile, and employees working overtime leave at nearly 3x the rate of those who don't. Job satisfaction and work-life balance follow similar patterns on their own where they have roughly 23% and 31% attrition respectively. But the real risk emerges when these factors overlap. Low-income employees who also work overtime have 58.5% attrition. Employees that show low satisfaction, poor work-life balance, and works overtime leave at 41% compared to just 6.6% for employees with none of these risk factors. In short, overtime acts as a multiplier that sharply worsens outcomes wherever it combines with low pay or low satisfaction, making it the highest-priority lever for reducing attrition, while compensation and satisfaction improvements matter most for employees already at the bottom of those scales.

8 & 9. Create meaningful visualizations to support your analysis. Build an interactive dashboard in Tableau or Power BI.



10. Summarize key insights from the analysis

Key Insights from the Employee Attrition Analysis are:

- i. **Overall attrition rate is 16.1%** — a meaningful but minority-class problem, meaning any predictive model built on this data needs to account for class imbalance.
- ii. **Overtime is the single strongest driver of attrition.** Employees working overtime leave at 30.5%, nearly 3x the rate of those who don't (10.4%). This is the clearest, most actionable signal in the entire dataset.
- iii. **Risk factors compound rather than act independently.** No single feature shows strong correlation with attrition by itself. But when high-risk features are put together, attrition correlates highly. Low-income employees who also work overtime hit 58.5% attrition, and employees facing low satisfaction, poor work-life balance and overtime together leave at 41%, compared to just 6.6% for employees with none of these factors.
- iv. **Low pay matters most at the bottom.** Attrition is highest in the lowest income quartile but flattens out above that. This looks like a pay-floor problem rather than a broad compensation problem across the whole workforce.
- v. **Early-career, junior employees are highest risk.** Younger age, lower job level, shorter tenure, and less time with a current manager all associate with higher attrition, pointing to a vulnerable "first few years" period rather than risk being evenly spread across tenure.
- vi. **Sales Representatives are the highest-risk role by far**, more than double the next-highest role, while Research Directors and Managers are the most stable (2.5–4.9%) — largely reflecting seniority and tenure differences.
- vii. **The dataset itself is clean and reliable** — no missing values, no duplicates, ready for modelling or dashboarding without additional preprocessing.

11. Provide business recommendations to reduce employee attrition.

Based on what the data showed, here are practical recommendations to reduce attrition, organized by the strongest signals found in the analysis.

i. **Address overtime and workload first — it's the biggest lever**

Employees working overtime leave at nearly 3x the rate of those who don't (30.5% vs 10.4%). This is the single strongest driver in the data, so it's the highest-priority fix.

- Audit which roles and teams are generating the most overtime, and investigate whether it reflects understaffing, poor workload distribution, or unrealistic deadlines.
- Consider caps on sustained overtime, additional headcount in overworked teams, or redistributing tasks rather than relying on the same people repeatedly.

- Track overtime hours as a leading indicator, not just attrition after the fact — by the time someone leaves, the workload problem has already been there for months.

ii. Focus retention efforts on early-tenure and junior employees

Attrition skews heavily toward younger employees, those newer to their role or manager, and those lower on the job ladder — all pointing to a vulnerable "first few years" period.

- Strengthen onboarding and early mentorship programs, since people who haven't yet built strong relationships with their manager or team are more likely to leave.
- Create clearer, visible promotion and growth paths for junior staff, since a lack of progress signals is often what pushes early-career employees out.
- Pay particular attention to the first 1–2 years of tenure with structured check-ins, since that's where the data shows the highest risk concentration.

iii. Prioritize Sales Representatives as a high-risk group

This role has by far the highest attrition rate (nearly 40%), significantly above every other role.

- Investigate this role specifically rather than applying department-wide fixes — compensation structure, quota pressure, and travel demands are worth reviewing in detail.
- Compare what Sales Reps experience versus more stable roles like Managers or Research Directors, since those roles differ mainly in seniority, income, and tenure, not necessarily in job content.
- Consider whether commission structures, career progression out of the role, or manager support differ meaningfully for this group.

iv. Review compensation for lower-income and lower-level employees

Lower monthly income correlates with higher attrition, though not overwhelmingly on its own — it compounds with the other factors above.

- Benchmark pay for lower job levels against market rates, especially in roles or departments already flagged as high-risk (Sales).
- Consider whether stock options or equity, which also showed a protective effect, could be extended further down the org chart as a retention tool.